

Code & Data Access

All code and data for this project are available at the following Google Drive folder:
<https://drive.google.com/drive/folders/1rjAOjDqIkIUfgY0JliLm-02H63cmerx?usp=sharing>

Setup & Execution Instructions

- **Notebook Location:**
 - File: SRAI_Course_Project_code.ipynb
 - Recommended to run using **Google Colab**
- **Required Keys:**

OpenAI API Key

Used for self-consistency verification via GPT models.

Get your key at: <https://platform.openai.com/account/api-keys>

Add it in the notebook with:

```
"""Generate a response using OpenAI API"""  
try:  
    # Replace with your actual API key  
    api_key = "open-ai-api-key"  
    client = OpenAI(api_key=api_key)
```

Hugging Face Token

Required to load transformer models (e.g., sentence-transformers).

When prompted, enter your token which can be found here:

<https://huggingface.co/settings/tokens>

Dataset Instructions

- The dataset is included in the provided Drive link.
- It contains both raw XML files and a preprocessed JSONL file.

Folder Structure:

```
ethical_rag_data/  
├── PMC000xxxxxx/      ← Raw XMLs from PubMed (optional)  
└── processed_articles.jsonl ← Preprocessed, ready-to-use file
```

- The raw PMC XML files were downloaded from the official **PubMed Central Open Access Bulk** archive: https://ftp.ncbi.nlm.nih.gov/pub/pmc/oa_bulk/

- The notebook will automatically parse and preprocess the XMLs to generate chunked documents.
However, `processed_articles.jsonl` is already provided in case you want to skip preprocessing.

Important Note:

If you change the dataset folder location or directory structure, update this line in the notebook:

with `open("data/processed/processed_articles.jsonl")`